

# PaperGrade

## Grade handwritten answer sheets — in the student's own language.

This started with my mother. She teaches **Marathi**, and every exam season she grades papers **by hand** — **3+ hours a day, 7 days an exam, 9 exams a year**.

That's **~189 hours a year — almost 8 full days** of unpaid work, on top of teaching. Millions of teachers face the exact same grind.

We're using Sarvam to put the benefits of AI into millions of average Indian households — solving a **real problem**, born from my mother's hardship.

### 1.47 million

schools in India (UDISE+ 2023-24)

### 248 million

students enrolled

### 22 languages

of instruction under NEP — the majority sat & graded **by hand**

👉 The judge test: **write two lines on paper, photograph on your phone, watch it grade — live.**

HOW IT WORKS · CLARITY & PRODUCTION-READY

# One photo in. A graded paper, a spoken explanation, and a QR — out.

01

Rubric → JSON

Teacher pastes a rubric; compiled once into structured criteria.

02

Photograph

Phone camera, in-browser. Whole batch at once.

03 · Vision

Extract per question

Handwriting read region-by-region.

04 · 105B

Grade + abstain

Marks + confidence; flags what it can't judge.

05

Review queue

Teacher confirms only the uncertain ones.

06 · Bulbul

Speak feedback

Per-student audio in their language.

07

QR on the paper

Student/parent scans → hears why marks were lost.

**The business unlock:** a Class-8 student has no login and no app. Every graded script gets a **QR the teacher staples on**. A parent hears, in their own language, exactly why their child lost two marks. No student account, no data collected.

**Five screens, hosted & live:** Setup · Capture · **Review console** · Student card · Class dashboard.

A judge opens a URL and it works in 90 seconds — the unguessable workspace link *is* the credential. No signup wall.

# Remove any one model, and the product breaks.

| MODEL              | ROLE — LOAD-BEARING, NOT AN ADD-ON                                                                                                             |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Sarvam Vision      | Reads the handwriting, per question region — the Indic-script OCR a Western model can't match.                                                 |
| Sarvam-105B        | Compiles the rubric & grades each answer against it — with confidence and the power to <b>abstain</b> .                                        |
| Bulbul             | Speaks each student's feedback in their own language — the delivery a text UI can't make.                                                      |
| Saaras (ASR)       | Teacher <i>speaks</i> the override reason in Hindi/Marathi — "तीन अंक, सूत्र गलत" — into the audit log. Teachers facing 40 scripts won't type. |
| Mayura / Translate | Rubric authored in English, feedback rendered in the student's language — one rubric, a multilingual classroom.                                |

Why not a generic LLM? Indic handwriting + in-language voice is precisely where sovereign Indian models win — and where a chatbot wrapper can't follow.

# The system knows what it doesn't know — and proves it.

## Confidence is measured, not self-reported.

Each crop is read twice (one pass rotated + rescaled); the **edit distance between the two transcripts** is the uncertainty. Disagreement, rendered:

जल का कथनांक कवथनाक सौ डिग्री सेल्सियस होता है।

Below threshold → routed to a human. A model that abstains demos better than one that always answers.

- **Per-question segmentation** — never a whole page for one grade
- **Rubric** → **JSON**, cached — grading is constrained, auditable output
- **Forced abstention** — `needs_human: true` instead of a guess

## QWK 0.96

quadratic weighted kappa vs. teacher marks · **n = 20**, reported honestly with sample size

## 5 of 6

students missed the same concept — **friction (83%)**. One query → the artefact a principal buys.

Live numbers from the working build

# A wedge into every regional-language classroom in India.

**Market.** State boards, coaching chains, ed-tech platforms.

Formative assessment is the wedge; board-exam evaluation is the TAM — across **1.47M schools · 248M students · 22 languages**. Coaching chains sell to **parents**, and a parent who hears *why* in their own language is a renewal.

**The moat US models can't copy.** Minors' answer scripts are protected data under India's DPDP Act. **Sovereign Indian models, Indian infrastructure (Mumbai / ap-south-1), data never leaves the country.**

Unlike an OCR-accuracy lead, the next Gemini release can't erase it.

## Roadmap

- Head-of-department concept analytics
- Org / roles / SSO for schools
- State-board evaluation pilots
- DPDP consent, teacher-of-record sign-off, SOC 2

Try it on your phone — right now

